

Forecasting Next-Day Stock Returns with Retail Attention: An Out-of-Sample Evaluation of WallStreetBets Mention Signals

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1. Abstract

This paper asks whether retail-investor attention, measured through the daily volume of ticker mentions on the Reddit forum r/WallStreetBets (WSB), carries economically meaningful, out-of-sample information about the direction of next-day stock returns. I assemble a panel of 25,031 daily observations covering twenty high-attention U.S. equities from 2021 through 2025, combining Yahoo Finance price data with historical WSB mention counts. From this panel I engineer five price-based features and five within-ticker-normalized mention features, and I train a gradient-boosted tree classifier (LightGBM) to predict the sign of the following day's return. Models are evaluated under a strict expanding walk-forward design with no lookahead, benchmarked against random forests, extremely randomized trees, penalized logistic regression, and a majority-class baseline, and assessed not only by statistical accuracy but by economic significance through information coefficients, quintile sorts, and a daily long-short portfolio. The central finding is sober. Out-of-sample discriminatory power is marginal: the best model achieves a mean test AUC of roughly 0.507, only slightly above the 0.500 coin-flip benchmark, and no model's information coefficient is statistically distinguishable from zero at conventional levels. After Platt scaling, the predicted-probability distribution widens enough to express occasional confidence, and a calibrated long-short portfolio earns a Sharpe ratio of 0.66 in 2025 but loses heavily in 2024 (Sharpe -0.63), so neither year supports a tradeable strategy. The signal that does appear concentrates in high-attention and high-volatility regimes, consistent with attention-driven price discovery. I interpret these results as a credible, mixed verdict: WSB mentions contain a faint, regime-dependent trace of predictive content, but one too weak and unstable, over this universe and sample, to constitute a reliable forecasting edge.

2. Introduction

The proposition that prices already reflect available information is the organizing idea of modern asset pricing (Fama, 1970), and a large body of out-of-sample evidence cautions that apparent return predictability often evaporates once researchers move beyond the in-sample fit (Welch & Goyal, 2008). At the same time, a parallel literature documents that investor attention is not uniformly distributed and that surges in attention move prices, at least temporarily. Barber and Odean (2008) show that individual investors are net buyers of attention-grabbing stocks; Da, Engelberg, and Gao (2011) find that direct measures of retail attention, such as Google search volume, predict short-horizon price pressure and subsequent reversal. The rise of social trading communities has made attention newly observable in real time.

Few venues illustrate this more vividly than r/WallStreetBets, the Reddit forum that coordinated the early-2021 episodes in GameStop (GME) and AMC and that continues to function as a barometer of retail enthusiasm. If the collective gaze of millions of retail participants conveys information, or even predictable order flow, then the daily count of how often a ticker is named on WSB ought to help forecast its near-term return. This paper tests that hypothesis directly and, importantly, honestly: it subjects the WSB mention signal to the same out-of-sample discipline that has humbled many richer predictors.

The contribution is threefold. First, I construct a clean daily panel that merges five years of price history with historical WSB mention counts for a curated universe of twenty tickers chosen for sustained mention coverage. Second, I design mention features that survive the enormous cross-sectional scale differences in raw mention counts, normalizing each ticker against its own history so that a spike in attention is measured relative to that name's baseline rather than against GME's hundreds of thousands of daily mentions. Third, and most importantly, I evaluate the resulting forecasts the way a practitioner would: with expanding walk-forward windows, probability calibration, and

explicitly economic metrics, including a tradeable long-short portfolio, rather than relying on a single in-sample accuracy number.

The results do not vindicate a naive enthusiasm for social-media signals, but neither do they dismiss them entirely. Predictive accuracy is close to chance and statistically insignificant on the full cross-section, yet the signal becomes detectable precisely where theory says it should be strongest: in high-attention, high-volatility names. I read this as a genuine, if modest, trace of attention-driven price discovery that is swamped by noise at the universe level. The remainder of the paper develops this argument. Section 3 reviews the relevant literature; Section 4 describes the data; Section 5 details the features, models, and evaluation design; Section 6 presents results; Section 7 discusses interpretation and limitations; and Section 8 concludes.

3. Background and Related Work

3.1 Retail Attention and Asset Prices

The behavioral-finance literature provides the theoretical motivation for an attention signal. Barber and Odean (2008) argue that because individual investors face a vast search problem when deciding what to buy, they disproportionately purchase stocks that have recently caught their attention, generating temporary buying pressure. Da, Engelberg, and Gao (2011) operationalize attention with Google search volume and document price run-ups followed by reversals, a pattern consistent with attention inducing transitory mispricing rather than revealing fundamentals. Tetlock (2007) shows that media tone, a related form of salience, forecasts short-horizon returns and trading volume. Within the specific context of online message boards, Antweiler and Frank (2004) find that the sheer volume of posts on financial forums predicts volatility and, more weakly, returns, while cautioning that much of the content is noise.

More recent work studies Reddit directly. Bradley, Hanousek, Jame, and Xiao (2021) examine the investment recommendations posted on r/WallStreetBets and find that, on average, the most upvoted ideas are followed by short-lived price impact rather than durable outperformance, echoing the attention-and-reversal pattern. Taken together, this body of work suggests that mention-based signals are most plausibly informative about short-horizon, attention-driven price movement, especially in retail-favored, high-volatility names, and least plausibly informative about fundamentals. That prior shapes the present design: I forecast at the one-day horizon and examine whether any signal concentrates in high-attention, high-volatility regimes.

3.2 Machine Learning for Return Prediction

On the modeling side, Gu, Kelly, and Xiu (2020) demonstrate that flexible machine-learning methods, particularly tree ensembles and neural networks, can extract economically meaningful predictive structure from large sets of

equity characteristics, while emphasizing that gains are concentrated and that rigorous out-of-sample protocols are essential. Gradient-boosted decision trees (Ke et al., 2017) are well suited to this problem because they capture nonlinearities and interactions without manual specification and are robust to features on disparate scales. A recurring theme in this literature, sharpened by López de Prado (2018) and by Bailey, Borwein, López de Prado, and Zhu (2014), is that the principal threat to credibility is backtest overfitting: with enough modeling freedom, an in-sample edge can almost always be manufactured. The defenses adopted here, walk-forward evaluation, a fixed feature set, and economic rather than purely statistical success criteria, follow directly from those warnings.

4. Data

4.1 Price Data

Daily open, high, low, close, adjusted close, and volume series were retrieved from Yahoo Finance for the period 2021 through 2025. Returns are computed from split-adjusted closing prices within each ticker. The one-day forward return defines the prediction target described in Section 5.2. All series are aligned to trading days and merged with the mention data on the (date, ticker) key.

4.2 WallStreetBets Mention Data

Retail attention is measured by the daily number of times each ticker is mentioned on r/WallStreetBets. Historical daily mention counts were collected for each year from 2021 through 2025, with each annual file covering roughly one hundred tickers ranked by mention volume. The defining empirical feature of these data is their extreme cross-sectional skew. On its busiest days, GameStop attracts on the order of 267,000 mentions, whereas a large but less retail-favored name such as JPMorgan typically draws only single-digit daily counts and peaks in the low hundreds. Raw mention counts therefore span more than four orders of magnitude across the universe, which

means they are dominated by which ticker is being measured rather than by whether that ticker is currently unusually discussed. Figure 1 illustrates the relationship between mentions and returns, and Figure 2 shows the characteristic spike pattern of mention activity for GameStop.

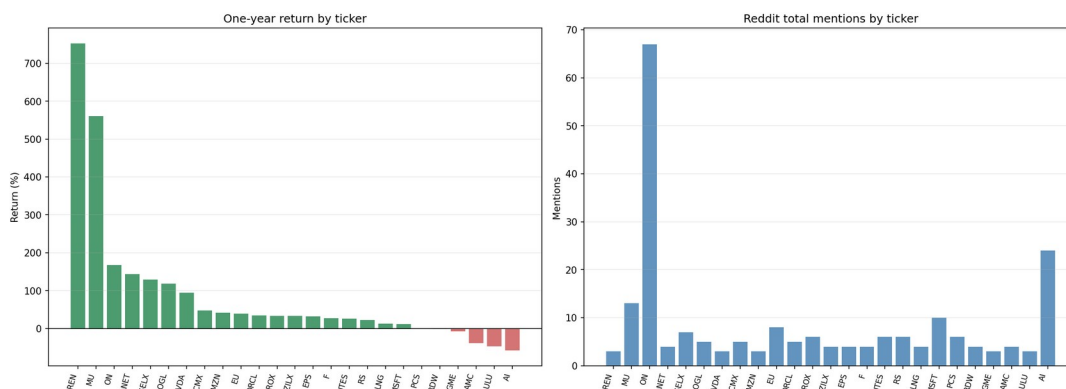


Figure 1. Daily WSB mentions and contemporaneous returns across the sample. Mention activity is highly concentrated and bursty.

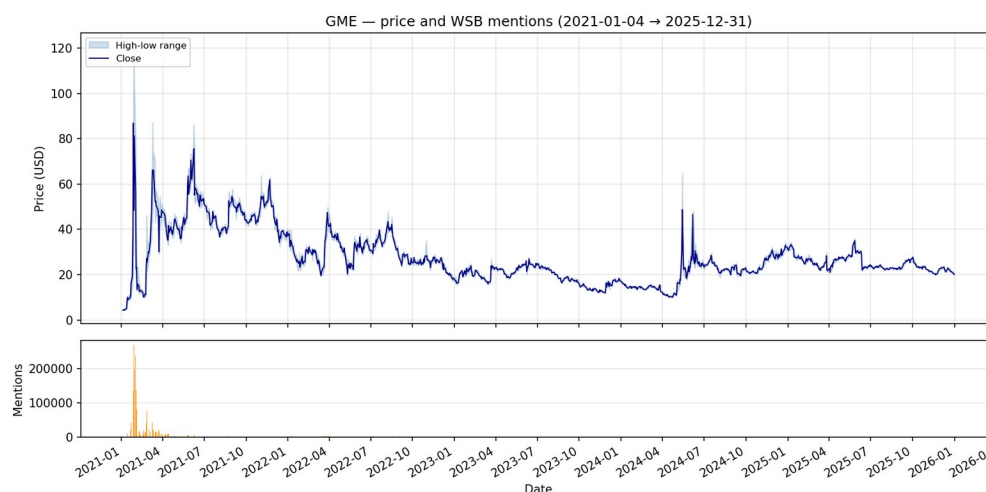


Figure 2. GameStop (GME) daily WSB mention volume, illustrating the episodic, heavy-tailed nature of retail attention.

4.3 Ticker Universe

To ensure that the mention signal is identified rather than swamped, I restrict the universe to twenty equities that combine sustained WSB coverage with meaningful mention volume. The selection requires at least four years of WSB coverage and excludes exchange-traded funds, broad indices, cryptocurrency proxies, and any ticker with fewer than two years of mention data. The

resulting universe spans large-cap technology anchors, high-attention semiconductor and electric-vehicle names, the core meme stocks, and a set of sector diversifiers, as summarized in Table 1. The final panel contains 25,031 daily observations across the twenty tickers and five years.

Category	Tickers
Large-cap anchors	AAPL, AMZN, MSFT
Semiconductor / EV	TSLA, NVDA, AMD, INTC, MU, TSM
Meme-stock core	GME, AMC
Retail-driven growth	PLTR, COIN, SNAP
Earnings-driven volatility	NFLX, LULU
Sector diversifiers	F, GM, BA, JPM

Table 1. The twenty-ticker universe, grouped by role in the cross-section.

5. Methodology

5.1 Feature Construction

All features are constructed so that they use only information available at the close of day T and are used to predict the return realized at the close of day $T+1$; there is no lookahead. Five features summarize price dynamics: the prior-day return, five-day and twenty-day price momentum, the rolling twenty-day standard deviation of daily returns as a volatility proxy, and the ratio of the day's volume to its trailing twenty-day average.

Five further features summarize mention activity, each engineered to neutralize the cross-ticker scale problem by normalizing a ticker against its own history. The first is the natural log of one plus the previous day's raw mention count, which compresses the heavy tail. The second is a mention-momentum term: the percentage change of a seven-day smoothed log-mention series relative to its value five trading days earlier. The third, abnormal mentions, is a rolling z-score of log-mentions against a thirty-day mean and standard deviation, capturing how unusual today's attention is for that specific name. The remaining two scale the abnormal-mention signal by realized volatility and by average traded volume, expressing attention per unit

of risk and per unit of liquidity, respectively. This within-ticker normalization was not the initial design; it was introduced after raw mention counts registered essentially zero importance, precisely because they were dominated by the cross-sectional scale differences documented in Section 4.

Table 2 lists the ten features. All mention features are lagged by one day relative to the target, and rolling statistics use only trailing windows, so the no-lookahead property holds by construction.

Feature	Description
return_1d	Prior-day return
momentum_5d	Five-day price momentum
momentum_20d	Twenty-day price momentum
volatility_20d	Rolling twenty-day return standard deviation
volume_ratio	Volume divided by trailing twenty-day average volume
mentions_log	Log of one plus yesterday's raw mention count
mentions_log_chg_5d	Change in seven-day smoothed log-mentions vs. five days ago
mentions_abnormal	Thirty-day rolling z-score of log-mentions
mentions_vol_scaled	Abnormal mentions divided by realized volatility
mentions_volume_scaled	Raw mentions divided by trailing average volume

Table 2. The ten predictive features. The first five are price-based; the last five summarize within-ticker-normalized retail attention.

5.2 Prediction Target

The target is the direction of the next-day return: it equals one if the following day's close exceeds the current close and zero otherwise. Framing the problem as binary classification rather than point forecasting reflects the modest ambition appropriate to the signal and lends itself to the probability-calibration and ranking diagnostics used later.

5.3 Models

The primary model is a LightGBM gradient-boosted tree classifier (Ke et al., 2017), configured conservatively to limit overfitting: three hundred trees, a learning rate of 0.03, a maximum of fifteen leaves, a minimum of twenty

samples per leaf, subsampling of both rows and columns at eighty percent, and L2 regularization, with early stopping after thirty rounds without validation improvement. To guard against the possibility that any result is an artifact of a single model class, I benchmark LightGBM against a random forest, an extremely randomized trees ensemble, logistic regression with L1 and with L2 penalties (each preceded by feature standardization), and a majority-class dummy classifier that always predicts the most frequent training label. All models share the identical feature set, target, and evaluation splits, and use a fixed random seed.

5.4 Walk-Forward Evaluation

Forecasts are evaluated under an expanding walk-forward scheme that respects the arrow of time. Two windows are used. In the first, models train on 2022, use 2023 only for early stopping and probability calibration, and are tested on 2024. In the second, the training set expands to 2022–2023, 2024 serves as the validation year, and 2025 is the held-out test year. Although the assembled panel begins in 2021, the learning pipeline is restricted to 2022 through 2025; the 2021 observations are excluded, and the earliest trading days of 2022 are dropped because the trailing windows needed for the rolling features are not yet populated. Critically, the validation year is reserved for early stopping and calibration only; it is never used to choose among models, so the test years remain genuinely out of sample. Each test year contributes roughly five thousand ticker-day observations.

Window	Train	Validation	Test
1	2022	2023	2024
2	2022–2023	2024	2025

Table 3. Expanding walk-forward windows. The validation year is used only for early stopping and calibration, never for model selection.

5.5 Probability Calibration

Tree ensembles trained on a near-random target tend to produce probabilities clustered tightly around the base rate, which makes them useless for

expressing confidence even if their ranking is informative. To address this, each fitted model is wrapped in Platt scaling (Platt, 1999): a sigmoid is fitted on the validation year to map raw scores onto better-calibrated probabilities, following the supervised-calibration approach of Niculescu-Mizil and Caruana (2005). Because calibration is a monotone transform, it leaves AUC essentially unchanged but can substantially improve probability spread and calibration error, as Section 6.3 documents.

5.6 Economic-Significance Metrics

Statistical accuracy alone is a poor guide to whether a signal is useful, so the evaluation emphasizes economic significance. I report the Spearman information coefficient (IC) between predicted probability and realized direction, both pooled and month by month; the Brier score and expected calibration error; mean realized returns across predicted-probability quintiles, summarized by the top-minus-bottom (Q5-Q1) spread; and a daily long-short portfolio that each day buys the two highest-ranked tickers and shorts the two lowest, reported through its annualized return, Sharpe ratio, maximum drawdown, and win rate. Finally, I compute conditional ICs within high- and low-attention and high- and low-volatility subsamples to test the behavioral prediction that any signal should concentrate in attention-driven, volatile names. Throughout, t-statistics and p-values accompany the IC and portfolio results so that economic magnitude is never divorced from statistical reliability. The full pipeline is implemented in Python with scikit-learn (Pedregosa et al., 2011) and LightGBM.

6. Results

6.1 Predictive Accuracy and Discrimination

Out-of-sample discrimination is marginal across every model. Table 4 reports test-year AUC for each classifier. LightGBM is the strongest performer, with AUCs of 0.504 in 2024 and 0.509 in 2025 for a mean of 0.507, but this edge over the 0.500 coin-flip benchmark is economically tiny and statistically

fragile. The tree ensembles cluster just below it, the penalized logistic models sit essentially at chance, and no model separates itself convincingly from the dummy baseline. Directional accuracy hovers near fifty percent throughout. Figure 3 visualizes the same comparison and makes the central message plain: every bar sits within a hair of the chance line.

Model	2024 AUC	2025 AUC	Mean AUC
LightGBM	0.504	0.509	0.507
Extra Trees	0.496	0.512	0.504
Random Forest	0.494	0.513	0.503
Logistic L1	0.497	0.498	0.498
Logistic L2	0.496	0.497	0.497
Dummy baseline	0.500	0.500	0.500

Table 4. Out-of-sample test AUC by model and year. LightGBM is the strongest, but all models are close to the 0.500 chance benchmark.

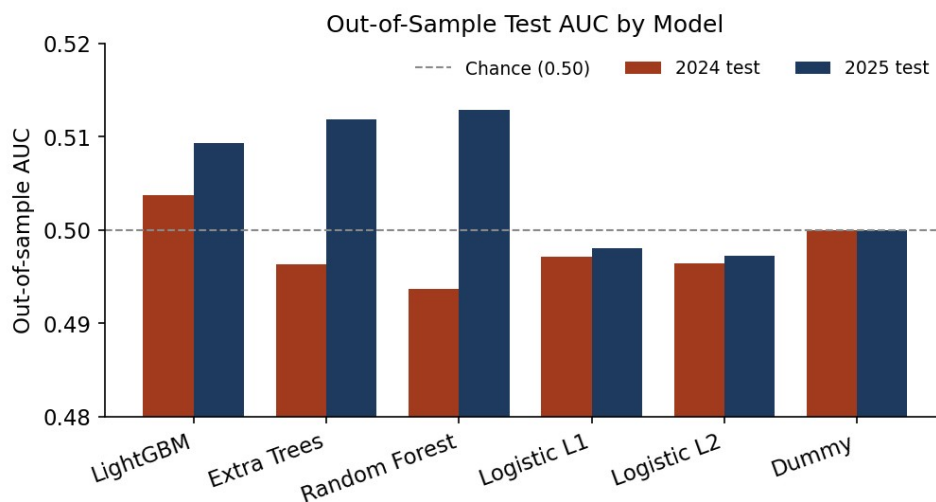


Figure 3. Out-of-sample test AUC by model for 2024 and 2025. The dashed line marks the 0.500 chance benchmark.

6.2 Feature Importance

Although discrimination is weak, the model does not ignore the mention features. Figure 4 shows LightGBM split-count importances for the 2022–2023 training window. The prior-day return and the five-day mention-momentum feature are the two highest-scoring inputs, tied at the top, with five-day price momentum close behind. All five mention features register non-

zero importance once within-ticker normalization is applied, in sharp contrast to the zero importance recorded by raw mention counts in earlier specifications. This is the clearest constructive result in the paper: properly normalized retail attention is informative enough that a disciplined model chooses to split on it as often as on price momentum, even if the resulting forecasts are only marginally better than chance.

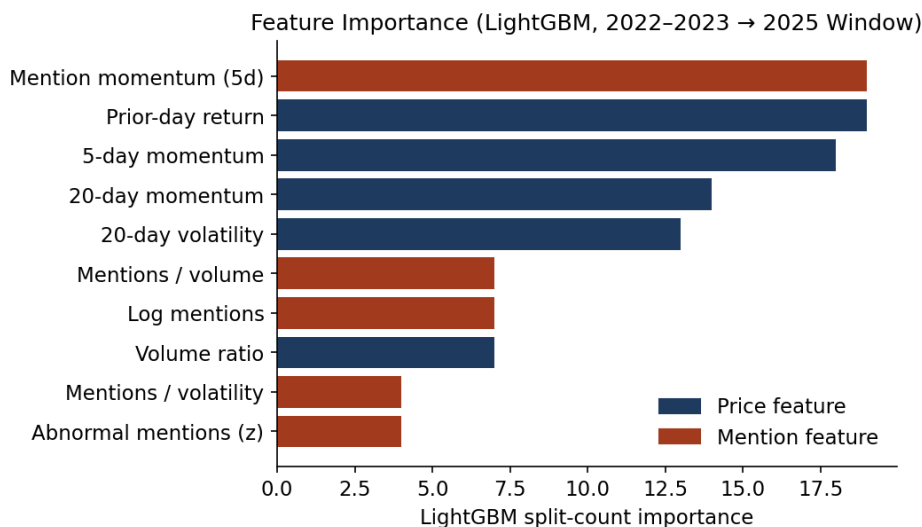


Figure 4. LightGBM split-count feature importance (2022-2023 window). Mention features (rust) are used as often as price features (navy); mention momentum ties prior-day return at the top.

6.3 Probability Calibration

Raw LightGBM probabilities are nearly degenerate: in 2024 they have a standard deviation of just 0.002 and never exceed 0.55, so the model cannot express any actionable confidence. Platt scaling corrects this. Table 5 shows that calibration widens the probability spread by roughly fivefold in 2024 and reduces the expected calibration error from 0.045 to 0.020; in 2025 it cuts calibration error from 0.011 to 0.002. After calibration, the model issues predictions above 0.55 for the first time, covering 7.8 percent of the 2024 cross-section. Brier scores barely move, as expected for a monotone transform, confirming that calibration improves the usefulness of the probabilities without manufacturing accuracy. Figure 5 summarizes the effect.

Metric	2024 Raw	2024 Cal.	2025 Raw	2025 Cal.
Std. of predicted probability	0.002	0.011	0.008	0.013
Expected calibration error	0.045	0.020	0.011	0.002
Brier score	0.252	0.250	0.250	0.250
Share of predictions > 0.55	0.0%	7.8%	0.0%	0.2%

Table 5. Effect of Platt scaling on probability quality for LightGBM. Calibration widens the probability distribution and lowers calibration error without changing the Brier score materially.

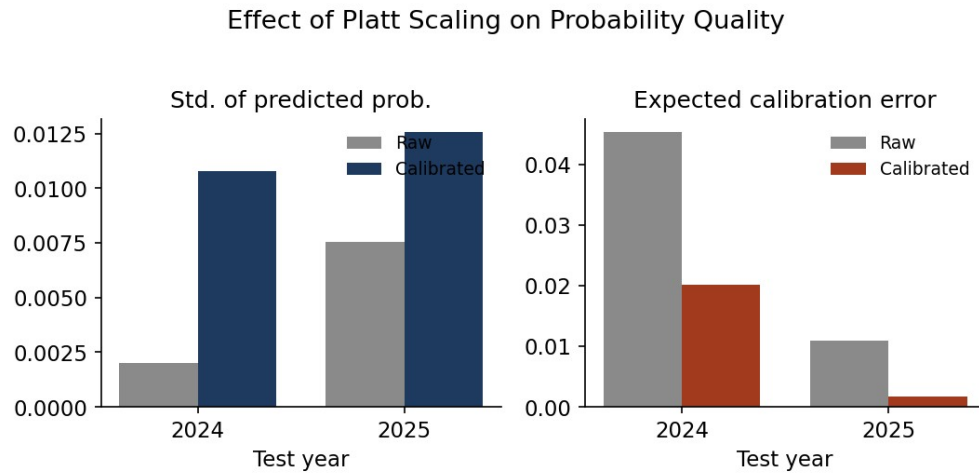


Figure 5. Probability spread and expected calibration error, before and after Platt scaling, by test year.

6.4 Signal Quality and the Information Coefficient

The Spearman information coefficient between the calibrated LightGBM probability and realized direction is 0.016 in 2025 with a p-value of 0.25, positive in sign but statistically indistinguishable from zero. The monthly IC averages about 0.010 and is positive in 58 percent of months, but it is lumpy: a single strong month (December 2025, with an IC of 0.12) accounts for much of the annual figure. The Q5-Q1 quintile spread is positive at 26.8 basis points per day in 2025, indicating that the highest-ranked names do modestly outperform the lowest-ranked, but the dispersion across the intervening quintiles is not cleanly monotone. The honest summary is that the ranking carries a faint positive signal whose statistical support is thin.

6.5 Long-Short Portfolio Performance

Translating the ranking into a daily top-two/bottom-two long-short portfolio crystallizes both the promise and the fragility of the signal. As Table 6 reports, the calibrated portfolio earns an annualized 30 percent with a Sharpe ratio of 0.66 in 2025, but loses an annualized 48 percent with a Sharpe of -0.63 in 2024, when the model's rankings were actively wrong. Maximum drawdowns are severe in both years, and the t-statistics, 0.66 in 2025 and -0.63 in 2024, indicate that neither result is statistically significant. Figure 6 plots the two equity curves: the 2025 portfolio grinds upward while the 2024 portfolio halves the initial dollar. A strategy that wins in one year and loses comparably in the adjacent year is not a tradeable edge; it is closer to noise with a wide variance.

Year	Ann. Return	Sharpe	Max Drawdown	Win Rate	t-stat
2024	-48%	-0.63	-69%	48.4%	-0.63
2025	+30%	+0.66	-36%	52.4%	+0.66

Table 6. Daily long-short portfolio (long top two, short bottom two) using calibrated LightGBM ranks. The strategy is profitable in 2025 and deeply unprofitable in 2024; neither year is statistically significant.

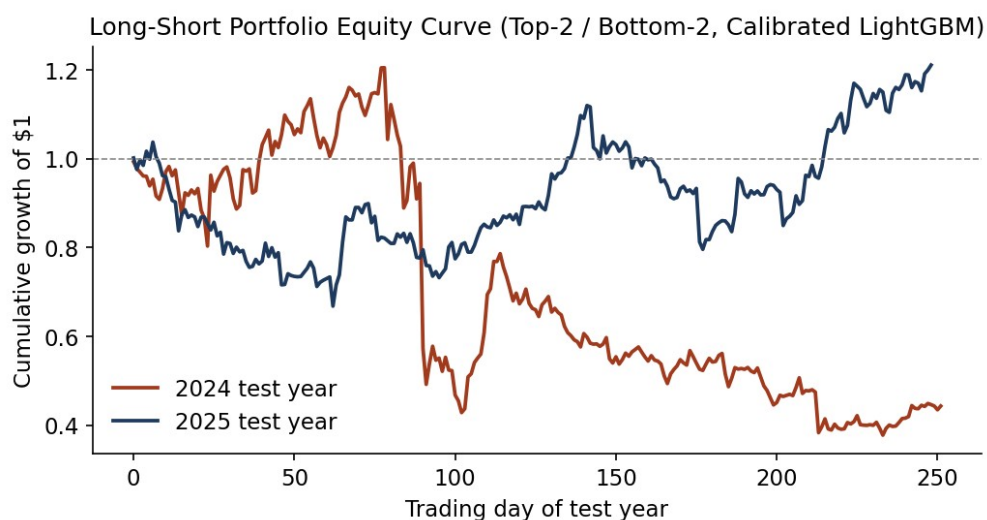


Figure 6. Cumulative growth of one dollar in the long-short portfolio for the 2024 and 2025 test years.

6.6 Conditional Performance by Regime

The most theoretically interesting result is conditional. When the sample is split by attention and by volatility, the information coefficient is positive and largest precisely in the regimes where behavioral theory predicts attention should matter. In high-abnormal-mention observations the IC is 0.033 ($p = 0.10$), against essentially zero in low-mention observations; in high-volatility observations it is 0.030 ($p = 0.14$), against slightly negative in calm names, as summarized in Table 7. None of these conditional ICs reaches conventional significance, but their consistent sign and concentration are exactly the pattern one would expect if WSB mentions proxy for attention-driven price discovery that is detectable only when attention and volatility are both elevated. The signal, in other words, is real enough to behave the way the theory says it should, yet too weak to clear a significance bar on this sample.

Regime	IC	p-value
High abnormal mentions	+0.033	0.10
Low abnormal mentions	-0.001	0.95
High volatility	+0.030	0.14
Low volatility	-0.006	0.78

Table 7. Conditional information coefficients (calibrated LightGBM, 2025 test year). The signal concentrates in high-attention and high-volatility regimes.

7. Discussion

7.1 Interpreting the Mixed Results

Three findings must be held together. First, at the level of the full cross-section, the WSB mention signal is close to useless for directional forecasting: AUCs sit near 0.50, ICs are insignificant, and the long-short portfolio is profitable in one test year and disastrous in the other. Second, the model nonetheless leans on the mention features, ranking mention momentum alongside the prior-day return in importance, which means the signal is not pure noise to the learning algorithm. Third, whatever signal exists concentrates in high-attention and high-volatility regimes. The reconciliation

that best fits this evidence is that retail attention carries a faint, transient, regime-dependent influence on next-day returns, consistent with the attention-and-reversal mechanism of Barber and Odean (2008) and Da et al. (2011), but that this influence is overwhelmed by idiosyncratic noise when averaged across calm, well-arbitraged, large-cap names. This is a more nuanced verdict than either techno-optimism or blanket dismissal, and it is consistent with the broader lesson that out-of-sample return predictability is real but small and unstable (Welch & Goyal, 2008; Gu et al., 2020).

It is worth stating plainly what would have constituted a positive result and was not observed: a test AUC reliably above, say, 0.53; an information coefficient significant at the five percent level; or a long-short portfolio with a positive Sharpe in both test years. None of these materialized. The discipline of walk-forward evaluation is doing exactly its job here, preventing an in-sample mirage from being mistaken for a strategy.

7.2 Limitations

Several limitations qualify these conclusions and motivate future work. The signal is built from mention counts alone; it contains no measure of sentiment polarity, so a flood of bearish posts and a flood of bullish posts look identical to the model. Incorporating tone scoring is the most promising single extension. The universe is also deliberately narrow at twenty tickers; the WSB data cover roughly one hundred names, and a broader cross-section would sharpen the conditional tests and improve the power of the IC and portfolio statistics, which are currently limited by having only two test years. The portfolio analysis abstracts from transaction costs, borrowing fees, and the considerable difficulty of shorting meme stocks, all of which would erode the already-insignificant 2025 result. The mention data are taken as given and may contain bot activity or measurement error. Finally, the daily horizon is one modeling choice among many; the attention literature suggests intraday or multi-day reversal horizons could behave differently. None of these caveats overturns the central finding, but each marks a direction in which the faint signal documented here might be amplified or shown to be illusory.

8. Conclusion

This paper subjected the popular intuition that r/WallStreetBets attention predicts stock returns to a deliberately demanding out-of-sample test. Using a five-year, twenty-ticker panel and a carefully engineered set of within-ticker-normalized mention features, a conservatively tuned gradient-boosted classifier achieved only marginal directional accuracy, with a mean test AUC of about 0.507 and no statistically significant information coefficient. A calibrated long-short portfolio earned a respectable Sharpe of 0.66 in 2025 but lost heavily in 2024, so the evidence does not support a tradeable strategy. Yet the signal is not nothing: the model relies on mention features as heavily as on price momentum, and the predictive content concentrates in exactly the high-attention, high-volatility regimes that behavioral theory implicates. The honest conclusion is that WallStreetBets mentions carry a faint, regime-dependent trace of information about next-day returns, too weak and unstable over this sample to forecast reliably, but real enough to behave as theory predicts and to justify the targeted extensions, sentiment scoring and a broader universe, that would test whether the trace can be sharpened into a signal.

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